

Q1 - 25 July - Shift 1

Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be :

- (A) 8 : 5 (B) 5 : 4
(C) 5 : 3 (D) 8 : 7

Space for your notes:

Q2 - 25 July - Shift 2

The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre.

The ratio of $\frac{B_2}{B_1}$ is :

- (A) $\frac{5}{2}$ (B) $\frac{25}{4}$
(C) $\frac{5}{4}$ (D) $\frac{25}{2}$

Space for your notes:

Q3 - 26 July - Shift 1

Questions

MathonGo

A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j})\text{T}$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j})\text{m/s}^2$, then the value of α will be

- (A) 3 (B) 6 (C) 12 (D) 2

Space for your notes:

Q4 - 26 July - Shift 1

B_x and B_y are the magnetic field at the centre of two coils of two coils X and Y respectively, each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_x and B_y is

- (A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) 4 : 1

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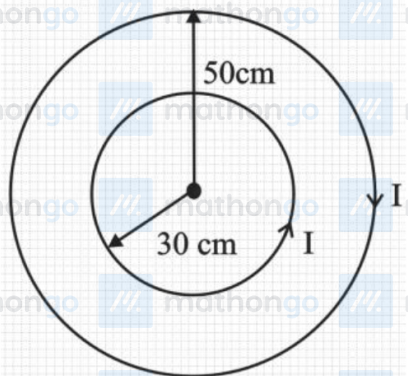
Q5 - 26 July - Shift 2

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Questions

MathonGo

Two concentric circular loops of radii $r_1=30$ cm and $r_2=50$ cm are placed in X-Y plane as shown in the figure. A current $I = 7$ A is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :



- (A) $\frac{7}{2} \hat{k} \text{ Am}^2$ (B) $-\frac{7}{2} \hat{k} \text{ Am}^2$
(C) $7 \hat{k} \text{ Am}^2$ (D) $-7 \hat{k} \text{ Am}^2$

Space for your notes:

Q6 - 27 July - Shift 2

A cyclotron is used to accelerate protons. If the operating magnetic field is 1.0 T and the radius of the cyclotron 'dees' is 60 cm, the kinetic energy of the accelerated protons in MeV will be :

[use $m_p = 1.6 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C]

- (A) 12 (B) 18
(C) 16 (D) 32

Space for your notes:

Q7 - 28 July - Shift 1

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The current sensitivity of a galvanometer can be increased by :

- (A) decreasing the number of turns
- (B) increasing the magnetic field
- (C) decreasing the area of the coil
- (D) decreasing the torsional constant of the spring

Choose the most appropriate answer from the options given below :

- (A) (B) and (C) only
- (B) (C) and (D) only
- (C) (A) and (C) only
- (D) (B) and (D) only

Q8 - 28 July - Shift 2

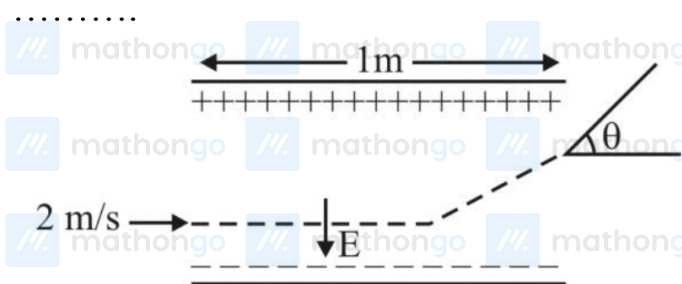
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Questions

MathonGo

A uniform electric field $E = (8m/e) \text{ V/m}$ is created between two parallel plates of length 1m as shown in figure, (where $m = \text{mass of electron}$ and $e = \text{charge of electron}$). An electron enters the field symmetrically between the plates with a speed of 2m/s . The angle of the deviation (θ) of the path of the electron as it comes out of the field will be

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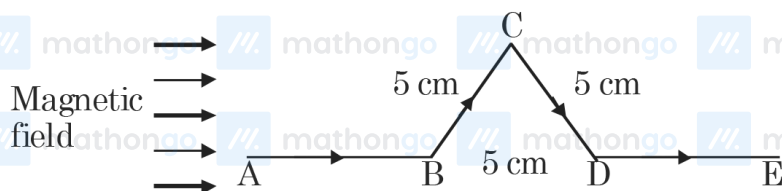


- (A) $\tan^{-1}(4)$ (B) $\tan^{-1}(2)$
 (C) $\tan^{-1}\left(\frac{1}{3}\right)$ (D) $\tan^{-1}(3)$

Q9 - 28 July - Shift 2

A triangular shaped wire carrying 10A current is placed in a uniform magnetic field of 0.5T , as shown in figure. The magnetic force on segment CD is (Given $BC = CD = BD = 5 \text{ cm}$).

Space for your notes:



- (A) 0.126 N (B) 0.312 N
 (C) 0.216 N (D) 0.245 N

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Q10 - 29 July - Shift 1

A closely wound circular coil of radius 5 cm produces a magnetic field of 37.68×10^{-4} T at its center. The current through the coil is _____ A.

[Given, number of turns in the coil is 100 and $\pi = 3.14$]

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Answer Key**Q1 (B)****Q2 (B)****Q3 (B)****Q4 (B)****Q5 (B)****Q6 (B)****Q7 (D)****Q8 (B)****Q9 (C)****Q10 (3)**

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Q1 (B)

$$\text{Radius of circular path } R = \frac{\sqrt{2mk}}{qB}$$

$$q = \frac{\sqrt{2mk}}{RB}$$

$$\frac{q_1}{q_2} = \sqrt{\frac{m_1}{m_2}} \times \frac{R_2}{R_1} = \sqrt{\frac{9}{4}} \times \frac{5}{6} = \frac{5}{4}$$

Q2 (B)

$$B = \frac{N\mu_0 i}{2R}$$

$$B_1 = \frac{N_1\mu_0 i}{2R_1}$$

$$\text{For } N_2 = 5$$

$$\text{Radius of coil} = R_2 = \frac{N_1 \times R_1}{N_2}$$

$$B_2 = \frac{N_2\mu_0 i}{R_2}$$

$$\frac{B_2}{B_1} = \frac{N_2}{N_1} \times \frac{R_1}{R_2} = \frac{N_2}{N_1} \times \frac{N_2}{N_1}; \quad \frac{B_2}{B_1} = \frac{25}{4}$$

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Q3 (B)

$$\text{As } \vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{a} = \frac{q}{m}(\vec{v} \times \vec{B})$$

So, $\vec{a}$ & $\vec{B}$ are $\perp$ to each other

$$\text{Hence, } \vec{a} \cdot \vec{B} = 0$$

$$(\alpha\hat{i} - 4\hat{j}) \cdot (2\hat{i} + 3\hat{j}) = 0$$

$$\alpha(2) + (-4)(3) = 0$$

$$\alpha = \frac{12}{2} \Rightarrow \alpha = 6$$

Q4 (B)

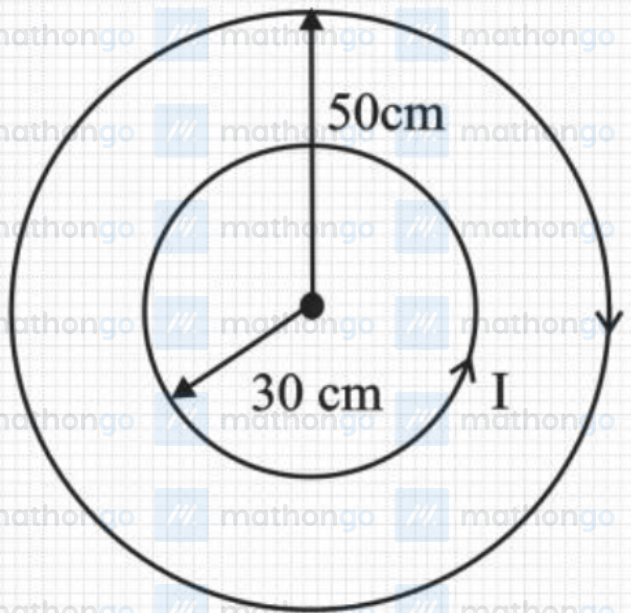
$$\text{At centre } B = N \left(\frac{\mu_0 i}{2R} \right)$$

$$B_x = 200 \left(\frac{\mu_0 i}{2 \times 20\text{cm}} \right)$$

$$B_y = 400 \left(\frac{\mu_0 i}{2 \times 20\text{cm}} \right)$$

$$\frac{B_x}{B_y} = \frac{1}{2}$$

Q5 (B)



Magnetic moment

$$\vec{M} = -i\pi(0.5)^2 \hat{k} + i\pi(0.3)^2 \hat{k}$$

$$\vec{M} = -7 \times \frac{22}{7} \left(\frac{25}{100} - \frac{9}{100} \right) \hat{k}$$

$$= -22 \left(\frac{16}{100} \right) \hat{k}$$

$$\vec{M} = -3.52 \hat{k} \text{ Am}^2$$

$$= -\frac{7}{2} \hat{k} \text{ Am}^2$$

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Q6 (B)

Kinetic energy of electron in cyclotron

$$= \left[\frac{q^2 B^2 r_0^2}{2m} \right]$$

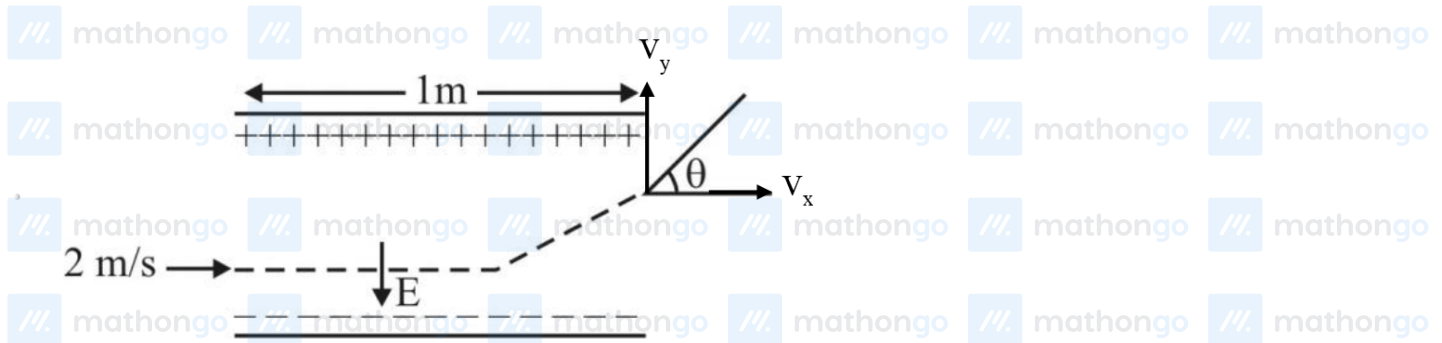
$$= 18 \text{ MeV}$$

Q7 (D)

$$i = \left(\frac{K}{NAB} \right) \theta$$

$$\therefore \frac{d\theta}{di} = \frac{NAB}{K}$$

Q8 (B)



$$a_y = \frac{F_y}{m} = \frac{e(E)}{m} = \frac{e \left(\frac{8m}{e} \right)}{m} = 8 \text{ m/s}^2$$

$$s_x = u_x t$$

$$1 = 2 \times t$$

$$t = \frac{1}{2} \text{ sec}$$

$$v_y = u_y + a_y t$$

$$v_y = 0 + 8 \times \frac{1}{2}$$

$$v_y = 4 \text{ m/s}$$

$$\tan \theta = \frac{v_y}{v_x} = \frac{4}{2} = 2 \Rightarrow \theta = \tan^{-1}(2)$$

$$\text{Q9 (C)}$$

$$F_M(\text{CD}) = B I \ell_{\text{eff}}$$

$$= 0.5 \times (10) \times (5 \sin 60 \times 10^{-2})$$

$$= 0.216 \text{ N}$$

$$\text{Q10 (3)}$$

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$$B_{\text{centre}} = \frac{N\mu_0 I}{2R}$$

$$37.68 \times 10^{-4} = \frac{100 \times 4\pi \times 10^{-7} \times I}{2 \times 5 \times 10^{-2}}$$

$$I = 3 \text{ A}$$